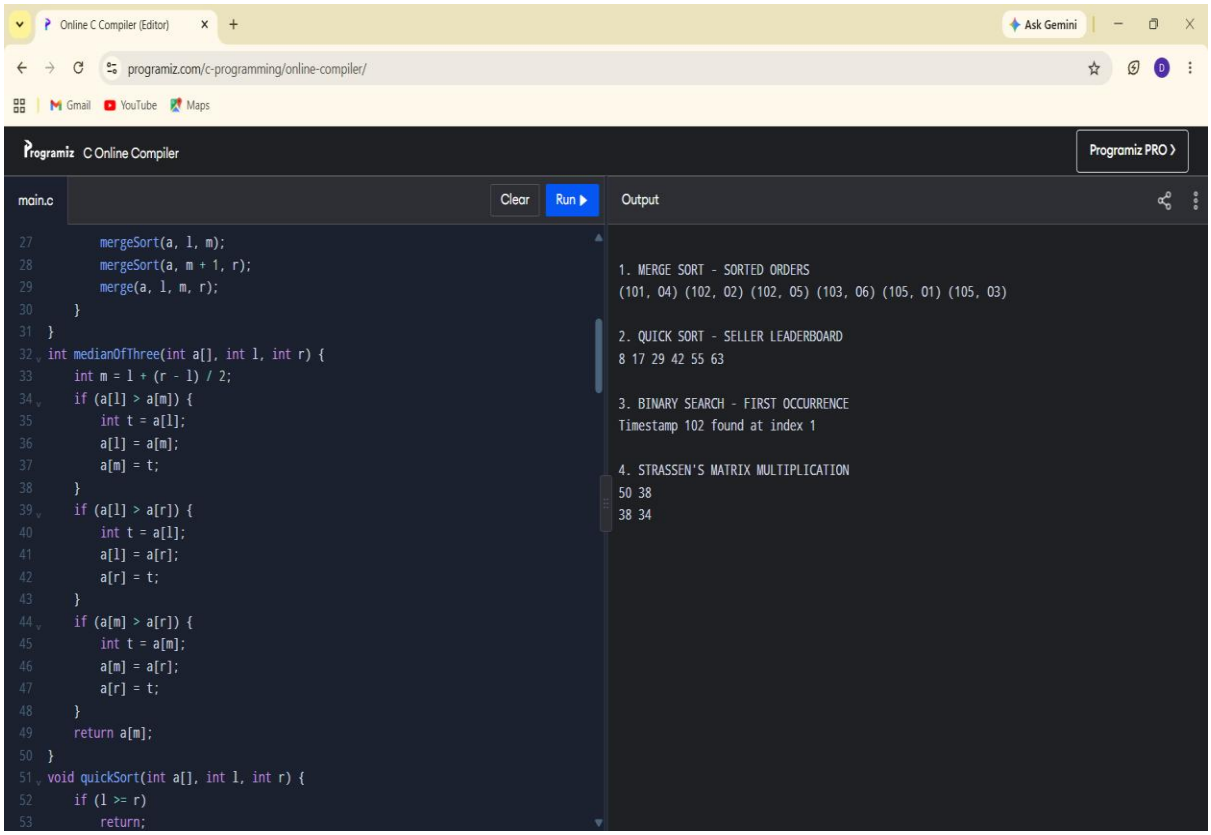
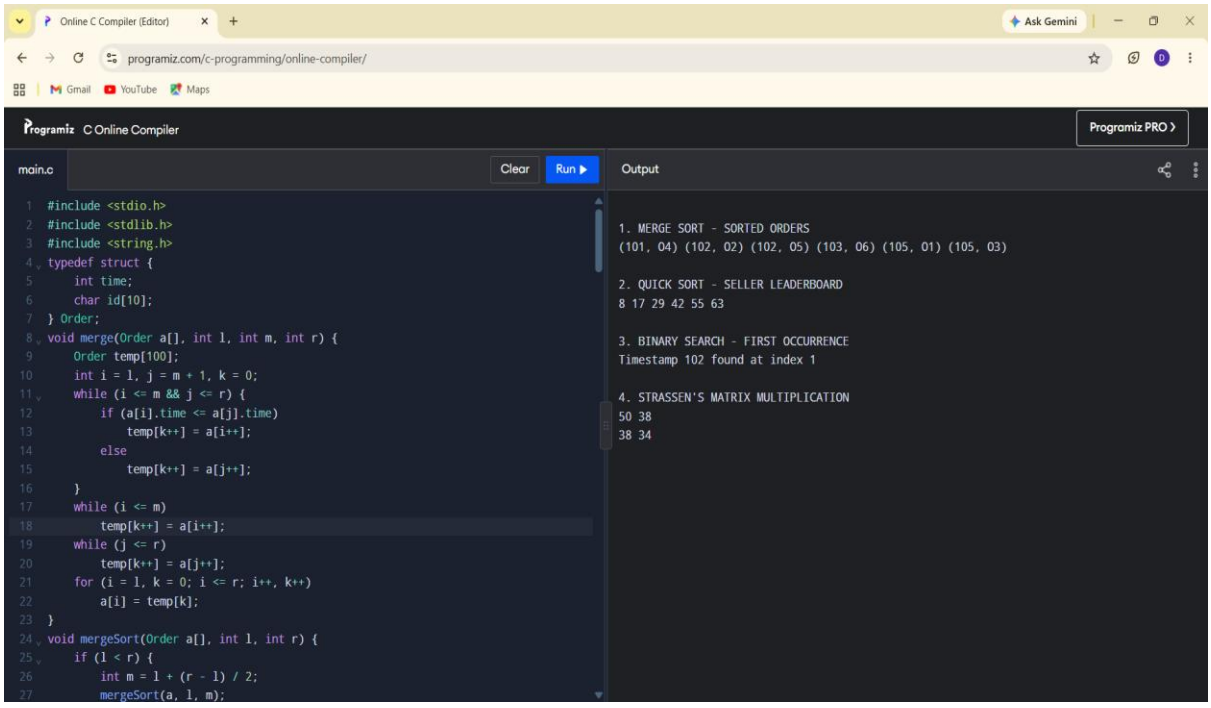
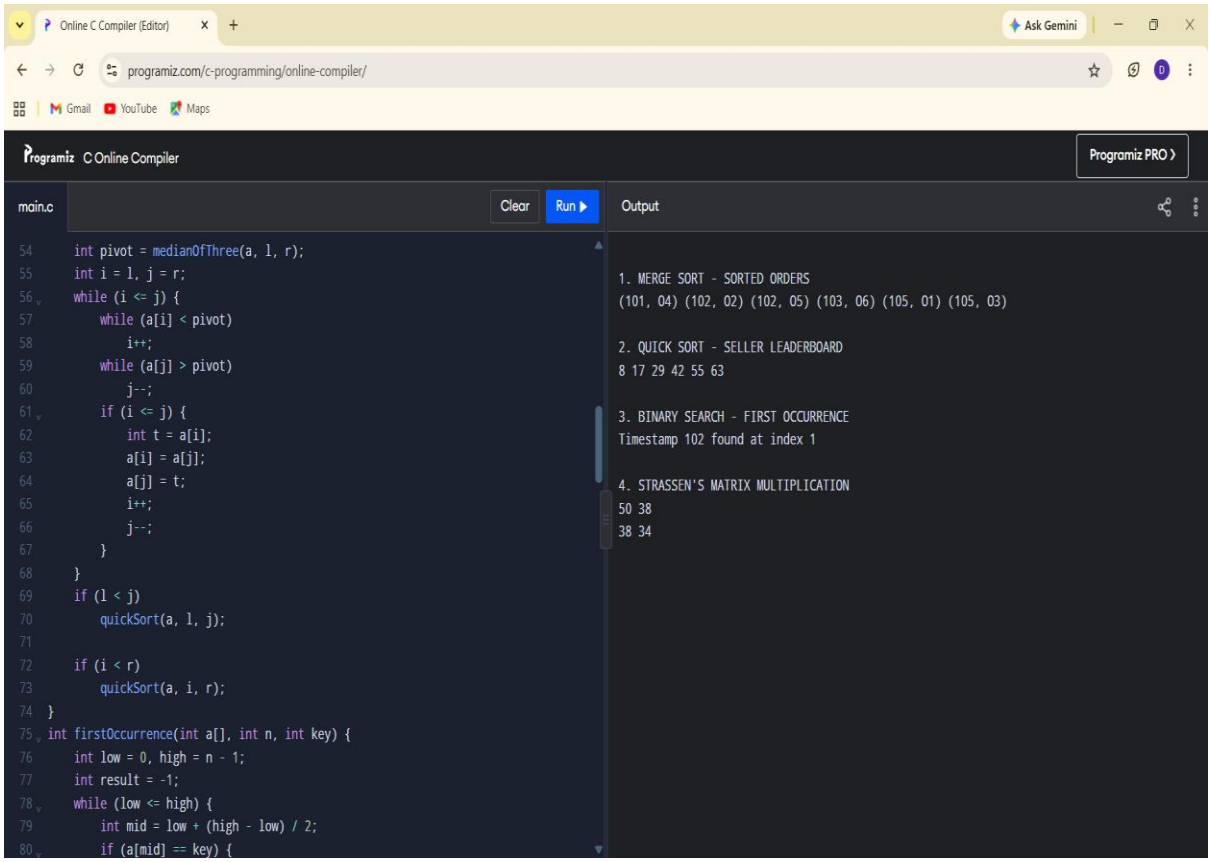
DESIGN AND ANALYSIS OF ALGORITHMS

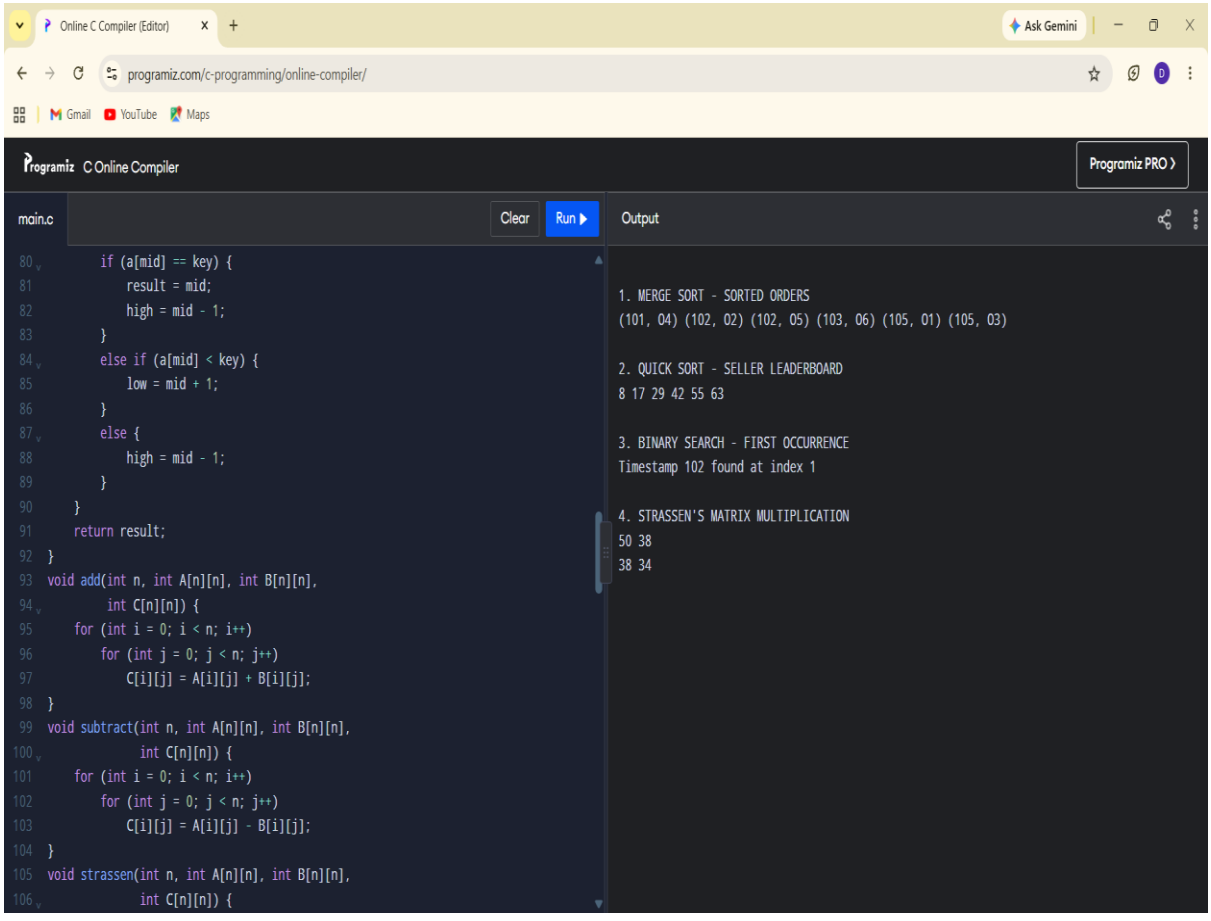
CO3 – CASE STUDY

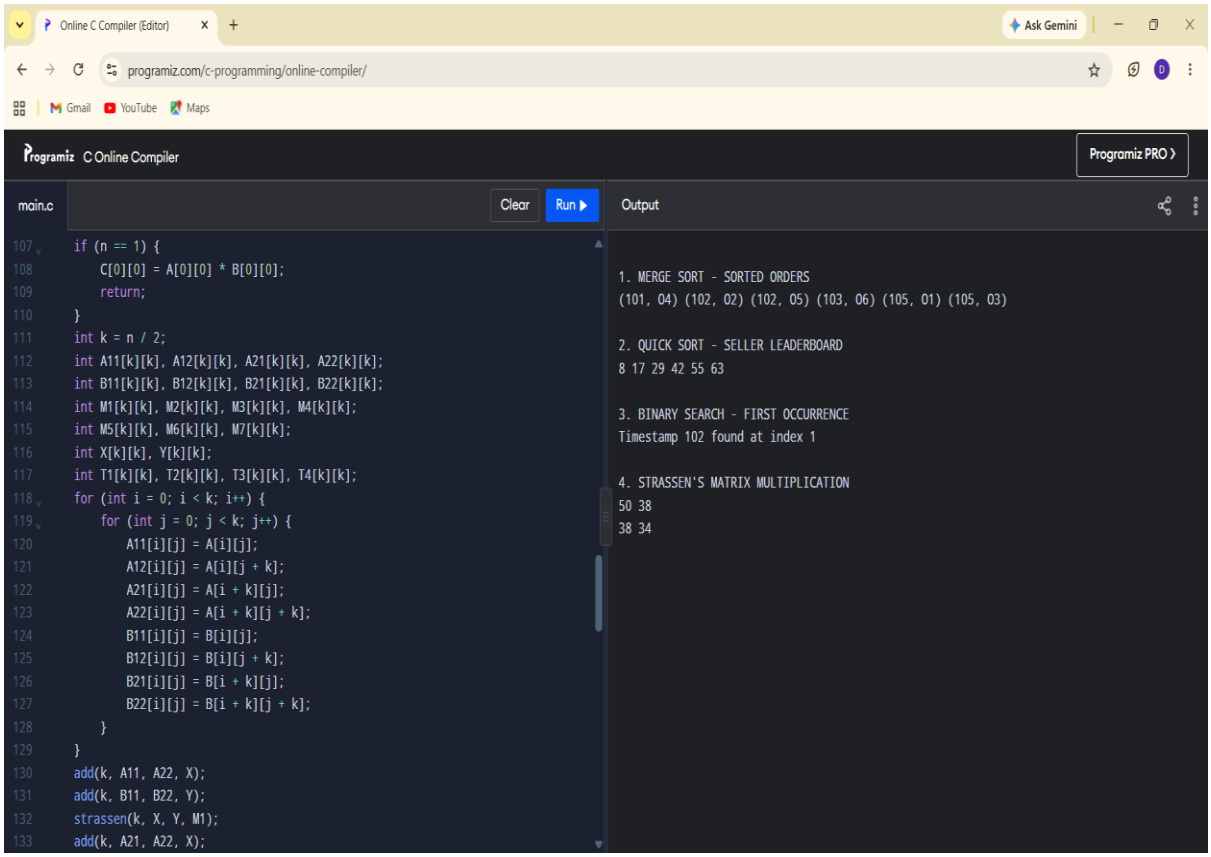
C PROGRAM IMPLEMENTATION

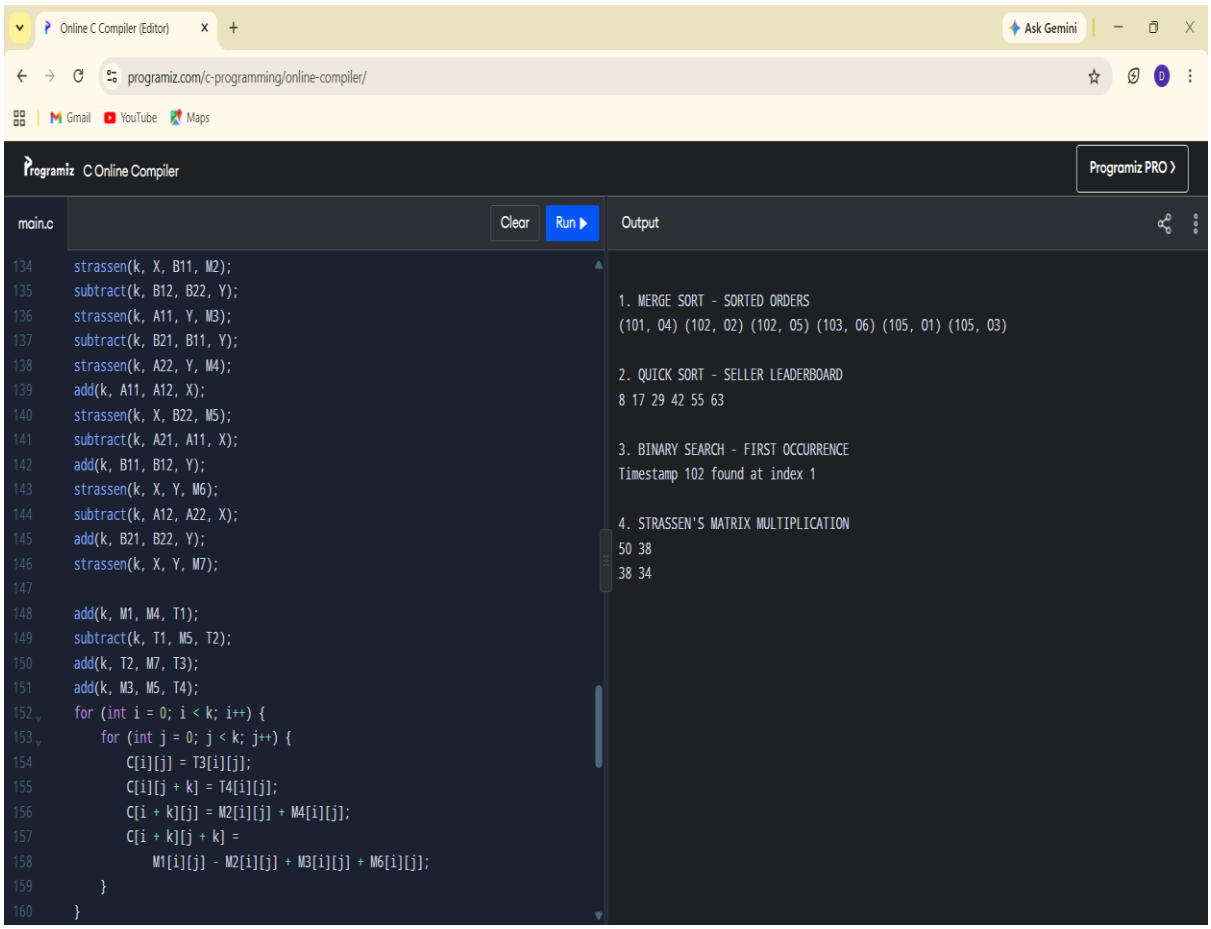


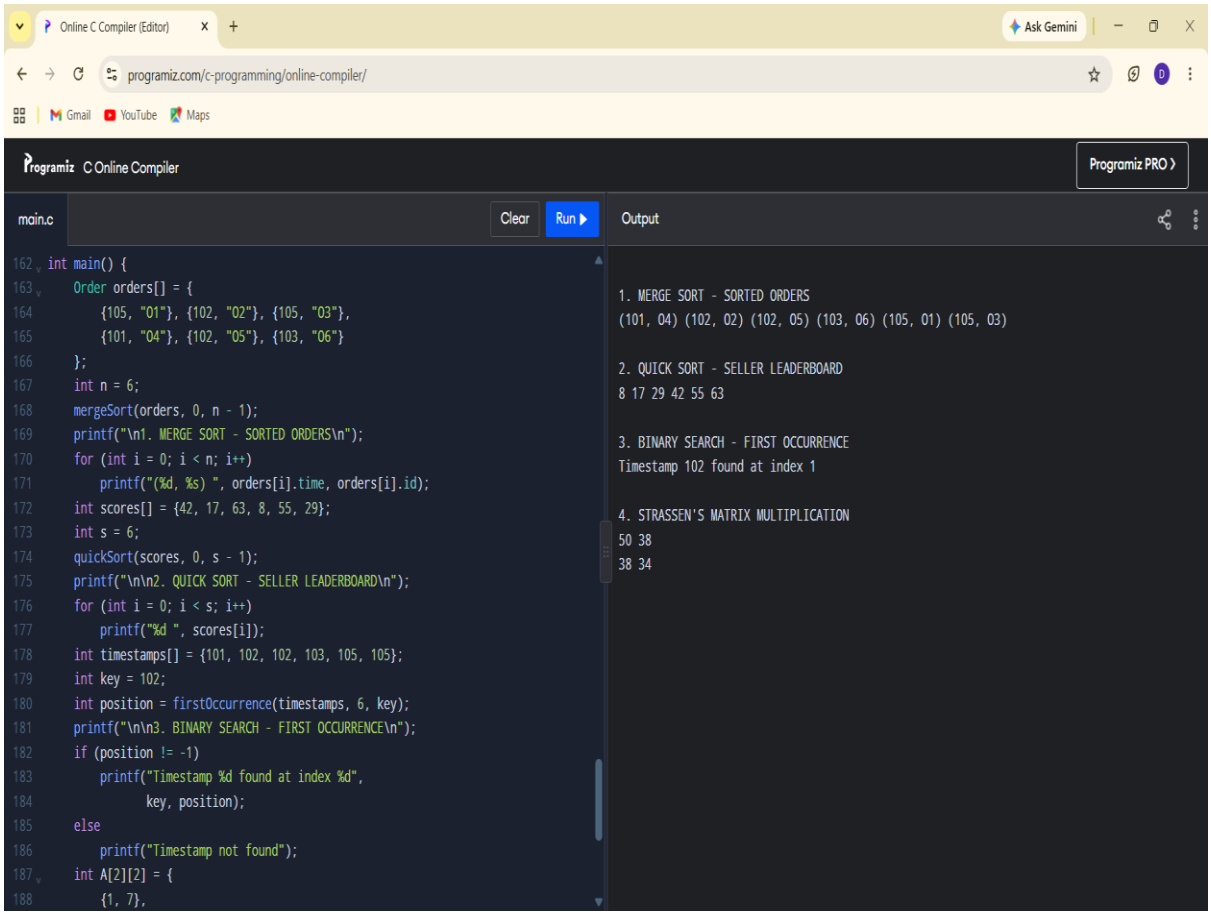


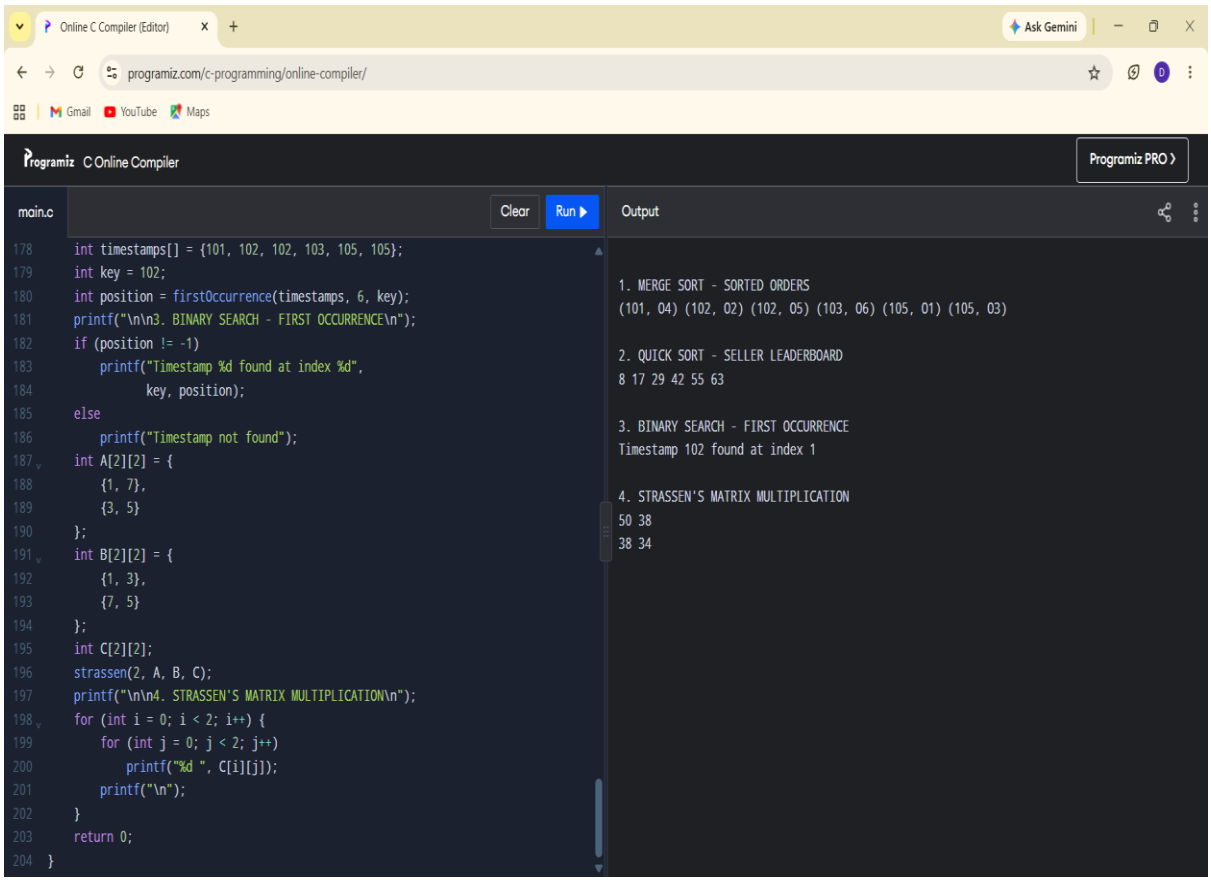

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main.c
Clear
Run
Output
int pivot = medianOfThree(a, l, r);
int i = l, j = r;
while (i <= j) {
while (a[i] < pivot)
i++;
while (a[j] > pivot)
j--;
if (i <= j) {
int t = a[i];
a[i] = a[j];
a[j] = t;
i++;
j--;
}
}
if (l < j)
quickSort(a, l, j);
if (i < r)
quickSort(a, i, r);
}
int firstOccurrence(int a[], int n, int key) {
int low = 0, high = n - 1;
int result = -1;
while (low <= high) {
int mid = low + (high - low) / 2;
if (a[mid] == key) {
1. MERGE SORT - SORTED ORDERS
(101, 04) (102, 02) (102, 05) (103, 06) (105, 01) (105, 03)
2. QUICK SORT - SELLER LEADERBOARD
8 17 29 42 55 63
3. BINARY SEARCH - FIRST OCCURRENCE
Timestamp 102 found at index 1
4. STRASSEN'S MATRIX MULTIPLICATION
50 38
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main.c
Clear
Run
Output
if (a[mid] == key) {
result = mid;
high = mid - 1;
}
else if (a[mid] < key) {
low = mid + 1;
}
else {
high = mid - 1;
}
}
return result;
}
void add(int n, int A[n][n], int B[n][n],
int C[n][n]) {
for (int i = 0; i < n; i++)
for (int j = 0; j < n; j++)
C[i][j] = A[i][j] + B[i][j];
}
void subtract(int n, int A[n][n], int B[n][n],
int C[n][n]) {
for (int i = 0; i < n; i++)
for (int j = 0; j < n; j++)
C[i][j] = A[i][j] - B[i][j];
}
void strassen(int n, int A[n][n], int B[n][n],
int C[n][n]) {
1. MERGE SORT - SORTED ORDERS
(101, 04) (102, 02) (102, 05) (103, 06) (105, 01) (105, 03)
2. QUICK SORT - SELLER LEADERBOARD
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main.c
Clear
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Output
if (n == 1) {
C[0][0] = A[0][0] * B[0][0];
return;
}
int k = n / 2;
int A11[k][k], A12[k][k], A21[k][k], A22[k][k];
int B11[k][k], B12[k][k], B21[k][k], B22[k][k];
int M1[k][k], M2[k][k], M3[k][k], M4[k][k];
int M5[k][k], M6[k][k], M7[k][k];
int X[k][k], Y[k][k];
int T1[k][k], T2[k][k], T3[k][k], T4[k][k];
for (int i = 0; i < k; i++) {
for (int j = 0; j < k; j++) {
A11[i][j] = A[i][j];
A12[i][j] = A[i][j + k];
A21[i][j] = A[i + k][j];
A22[i][j] = A[i + k][j + k];
B11[i][j] = B[i][j];
B12[i][j] = B[i][j + k];
B21[i][j] = B[i + k][j];
B22[i][j] = B[i + k][j + k];
}
}
add(k, A11, A22, X);
add(k, B11, B22, Y);
strassen(k, X, Y, M1);
add(k, A21, A22, X);
1. MERGE SORT - SORTED ORDERS
(101, 04) (102, 02) (102, 05) (103, 06) (105, 01) (105, 03)
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main.c
Clear
Run
Output
strassen(k, X, B11, M2);
subtract(k, B12, B22, Y);
strassen(k, A11, Y, M3);
subtract(k, B21, B11, Y);
strassen(k, A22, Y, M4);
add(k, A11, A12, X);
strassen(k, X, B22, M5);
subtract(k, A21, A11, X);
add(k, B11, B12, Y);
strassen(k, X, Y, M6);
subtract(k, A12, A22, X);
add(k, B21, B22, Y);
strassen(k, X, Y, M7);
add(k, M1, M4, T1);
subtract(k, T1, M5, T2);
add(k, T2, M7, T3);
add(k, M3, M5, T4);
for (int i = 0; i < k; i++) {
for (int j = 0; j < k; j++) {
C[i][j] = T3[i][j];
C[i][j + k] = T4[i][j];
C[i + k][j] = M2[i][j] + M4[i][j];
C[i + k][j + k] =
M1[i][j] - M2[i][j] + M3[i][j] + M6[i][j];
}
}
1. MERGE SORT - SORTED ORDERS
(101, 04) (102, 02) (102, 05) (103, 06) (105, 01) (105, 03)
2. QUICK SORT - SELLER LEADERBOARD
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main.c
Clear
Run
Output
int main() {
Order orders[] = {
{105, "01"}, {102, "02"}, {105, "03"},
{101, "04"}, {102, "05"}, {103, "06"}
};
int n = 6;
mergeSort(orders, 0, n - 1);
printf("\n1. MERGE SORT - SORTED ORDERS\n");
for (int i = 0; i < n; i++)
printf("(%d, %s) ", orders[i].time, orders[i].id);
int scores[] = {42, 17, 63, 8, 55, 29};
int s = 6;
quickSort(scores, 0, s - 1);
printf("\n\n2. QUICK SORT - SELLER LEADERBOARD\n");
for (int i = 0; i < s; i++)
printf("%d ", scores[i]);
int timestamps[] = {101, 102, 102, 103, 105, 105};
int key = 102;
int position = firstOccurrence(timestamps, 6, key);
printf("\n\n3. BINARY SEARCH - FIRST OCCURRENCE\n");
if (position != -1)
printf("Timestamp %d found at index %d",
key, position);
else
printf("Timestamp not found");
int A[2][2] = {
{1, 7},
1. MERGE SORT - SORTED ORDERS
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2. QUICK SORT - SELLER LEADERBOARD
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3. BINARY SEARCH - FIRST OCCURRENCE
Timestamp 102 found at index 1
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main.c
Clear
Run
Output
int timestamps[] = {101, 102, 102, 103, 105, 105};
int key = 102;
int position = firstOccurrence(timestamps, 6, key);
printf("\n\n3. BINARY SEARCH - FIRST OCCURRENCE\n");
if (position != -1)
printf("Timestamp %d found at index %d",
key, position);
else
printf("Timestamp not found");
int A[2][2] = {
{1, 7},
{3, 5}
};
int B[2][2] = {
{1, 3},
{7, 5}
};
int C[2][2];
strassen(2, A, B, C);
printf("\n\n4. STRASSEN'S MATRIX MULTIPLICATION\n");
for (int i = 0; i < 2; i++) {
for (int j = 0; j < 2; j++)
printf("%d ", C[i][j]);
printf("\n");
}
return 0;
}
1. MERGE SORT - SORTED ORDERS
(101, 04) (102, 02) (102, 05) (103, 06) (105, 01) (105, 03)
2. QUICK SORT - SELLER LEADERBOARD
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3. BINARY SEARCH - FIRST OCCURRENCE
Timestamp 102 found at index 1
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